JONATHAN ALEXANDER YATES

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EDUCATION

Purdue University — West Lafayette, IN Expected May 2029
Bachelor of Science in Aerospace Engineering, Specialization in Autonomy and Control
GPA: 3.78 | **Honors:** Dean's List (2025, 2026), Semester Honors (2025, 2026)

TECHNICAL SKILLS

CAD & Analysis: Autodesk Fusion 360 (Certified CAD/CAM), Siemens NX (CAD & FEM), GD&T, Blueprint Reading
Controls & Software: Rockwell Studio 5000, RSLogix 500, FactoryTalk Optix Studio & View, GE iFIX, MATLAB, Python, C, C++, ROS
Fabrication & Hardware: CNC Mill CAM, Manual Lathe, GMAW/SMAW Welding (CTE Certified), Soldering, 3D Printing
Systems & Testing: Industrial Ethernet/IP Configuration, Avionics Hardware Packaging, Sensor Interfacing, Instrument Calibration

ENGINEERING & RESEARCH EXPERIENCE

NSG Pilkington — Laurinburg, NC Summer 2026
Electrical & Systems Engineering Intern

- Programmed and troubleshot plant automation logic and safety interlocks utilizing Rockwell RSLogix 500 and Studio 5000 PLCs.
- Developed operator human-machine interfaces (HMI) and diagnostic alarm dashboards within Rockwell FactoryTalk Optix and GE iFIX.
- Configured industrial Ethernet/IP networks, subnet masks, and dedicated servers for machine-vision cameras and remote I/O drops.

Purdue MARS Lab - Guardian Rover — Purdue University Fall 2026 - Present
Undergraduate Researcher - Controls & Mechanical Design

- Contribute to development of Guardian rover control systems using C/C++ for microcontrollers and Python for control scripts.
- Support development and testing of ROS-based rover software and integration between microcontrollers and hardware.
- Create and modify 3D CAD models and assemblies to support rover mechanical design and system integration.

Smithsonian MMT Observatory — Mount Hopkins, AZ Summer 2024
Engineering Shadow

- Assisted lead staff engineers with mechanical maintenance, disassembly, and precision optical calibration on telescope assemblies.

Thunderbolt Welding — Midland, NC 2022 - 2024
Fabricator & Welder

- Fabricated custom structural steel weldments, utility trailers, and fixtures adhering strictly to engineering blueprints and tolerances.

ENGINEERING PROJECTS & LEADERSHIP

Purdue Space Program Active Controls - TOAD Lander Rocket — Purdue University Aug 2025 - Present
Structures Team Lead & Avionics Hardware Contributor

- Direct structural design, load path analysis, and verification for an actively stabilized thrust-vectoring testbed rocket.
- Generate CNC toolpaths in Fusion 360 CAM and fabricate motor mounts and brackets via CNC mills, lathes, and welding.
- Coordinate with avionics teammates to integrate flight computers, thrust-vectoring servos, and harness packaging into chassis.

University of Arizona Solar Oven Engineering Challenge — Tucson, AZ 2024
Project Team Member & Challenge Winner (1st Place)

- Engineered, modeled, and tested a parabolic solar thermal oven, achieving 1st place in overall thermal concentration and structural design.

Purdue Scuba Club — Purdue University Aug 2025 - Present
Vice President

- Direct organizational operations, manage club technical gear inspections, and coordinate logistics for dive excursions and safety training.

CERTIFICATIONS

- **Autodesk Certified User:** Fusion 360 CAD/CAM Design & Modeling
- **OSHA-10 General Industry Safety Certification:** Hazard recognition & safety protocols
- **CTE Welding & Fabrication Certification:** Blueprint reading, layout, & structural welding

SSI Advanced Open Water Diver: Underwater navigation, gear rigging, & field safety